

Experiment-9

```
Enter number of documents: 3
Enter document 1: The judgement was declared yesterday
Enter category 1: legal
Enter document 2: AI is transforming the world
Enter category 2: technology
Enter document 3: MLOps project is finalized
Enter category 3: project

Actual Labels:          ['legal', 'technology', 'project']
Rule-Based Predictions: ['agreement', 'agreement', 'agreement']
MaxEnt Predictions:     ['legal', 'technology', 'project']

Rule-Based Accuracy: 0.0
Maximum Entropy Accuracy: 1.0
```

Experiment -10

```
Enter number of posts: 3
Enter post 1: The restaurant interior is very beautiful
Enter post 2: cost is high
Enter post 3: food tastes good
Enter number of clusters: 3
```

Cluster Results:

```
Post: The restaurant interior is very beautiful
Cluster: 2
```

```
Post: cost is high
Cluster: 0
```

```
Post: food tastes good
Cluster: 1
```

Important Keywords:

```
Cluster 0
high
cost high
cost
tastes good
tastes
```

```
Cluster 1
tastes good
tastes
good
food tastes
food
```

```
Cluster 2
restaurant interior
restaurant
interior beautiful
interior
beautiful
```

Marketing Insight:

```
Similar customer opinions are grouped together.
Clusters help identify product trends and issues.
```